PHY 303- Electrodynamics-End Semester Examination

Please write short and concise answers. Illegible answers will not be graded.
Maximum marks =50, Time = 3 hours

1. (a) Two uniform infinite sheets of surface charge densities $+\sigma$ and $-\sigma$ intersect at right angles. Find the magnitude and direction of the electric field everywhere and draw the electric field lines. [5]

   (b) The space between two concentric shells of radii and b ($a < b$) is filled with an inhomogeneous material of dielectric constant

   $$\epsilon = \frac{\epsilon_0}{1 + Kr}$$

   where ϵ_0 and K are constants. A charge q is placed at the inner surface while the outer surface is grounded. Find

   i. The displacement and the electric field in the region $a < r < b$

   ii. The capacitance of the device.

   iii. The polarization charge density in the region $a < r < b$. [2,2,1]

2. (a) A current distribution gives rise to the magnetic vector potential

   $$\vec{A}(x, y, z) = x^2 y\hat{i} + y^2 x\hat{j} - xyz\hat{k}$$

   Find the corresponding magnetic field $\vec{B}$ at $(-1, 2, 5)$. [3]

   (b) The magnetic vector potential due to a small loop carrying current I may be expressed in the form

   $$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{I dl}{R},$$

   where $R = \sqrt{r^2 + r'^2 - 2rr' \cos\theta}$.

   Expand the expression in terms of Legendre polynomials at large distance (assuming Coulomb Gauge $\vec{\nabla}.\vec{A} = 0$). [7]

3. (a) In a region of empty space, the magnetic field is given by

   $$\vec{B} = B_0 e^{ax} \sin(ky - \omega t)\hat{x}.$$

   Calculate $\vec{E}$. [5]

   (b) Given a plane polarised electric wave,

   $$\vec{E} = \vec{E_0} exp\left\{i\omega\left[t - \frac{n}{c}(\vec{k}.\vec{r})\right]\right\}$$

   derive from Maxwell equations, relations between $\vec{E}$, $\vec{k}$ and $\vec{H}$ fields. Find the velocity of propagation and the refractive index . [3,1,1]

4. (a) Compute the intensity of the standing electromagnetic wave [5]

$$\begin{aligned} E_y(x,t) &= 2E_0 \cos kx \cos \omega t \\ B_z(x,t) &= 2B_0 \sin kx \sin \omega t. \end{aligned}$$

(b) A stationary observer measures the Earth's magnetic field to be $30\mu T$. What will be the magnetic and electric field (if any) measured by an observer moving past with a velocity 250 m/s perpendicular to the direction of the magnetic field (Hint: choose axes appropriately!). [5]

5. The scalar and vector potential of a dipole are given by

$$\begin{aligned} \Phi(r,\theta,t) &= -\frac{p_0\omega}{4\pi\epsilon_0 c}\frac{\cos\theta}{r}\sin[\omega(t-r/c)] \\ \vec{A}(r,\theta,t) &= -\frac{\mu_0 p_0 \omega}{4\pi r}\sin[\omega(t-r/c)]\hat{z} \end{aligned}$$

Compute

(a) The electric and magnetic fields (assuming Lorenz gauge).

(b) The Poynting vector.

(c) Intensity and the time averaged power radiated. [4,2,4]

Useful Formulae

Legendre Polynomials:

$$\begin{aligned} P_0(x) &= 0 \\ P_1(x) &= x \\ P_2(x) &= \frac{1}{2}(3x^2-1) \\ \int_{-1}^{1} P_m(x)P_n(x)dx &= \frac{2}{2n+1}\delta_{mn} \end{aligned}$$

Divergence and Curl in spherical coordinates:

$$\frac{1}{r^2}\frac{\partial(r^2A_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}$$

$$\begin{aligned} &\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(A_\varphi\sin\theta) - \frac{\partial A_\theta}{\partial\varphi}\right)\hat{\mathbf{r}} \\ &+\frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\varphi} - \frac{\partial}{\partial r}(rA_\varphi)\right)\hat{\boldsymbol{\theta}} \\ &+\frac{1}{r}\left(\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right)\hat{\boldsymbol{\varphi}} \end{aligned}$$

Electromagnetic field tensor:

$$F^{ij} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$